Jacob Rodgers

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Education

Oregon State University — Corvallis, OR
M.S. Physics — September 2024–June 2026 *(expected)*
Graduate Certificate in College and University Teaching

Michigan State University — East Lansing, MI
B.A. Physics — April 2024
Minor in Computational Mathematics, Science and Engineering

Teaching

Oregon State University

Graduate Teaching Assistant
Experience spans studio, recitation, and fully online (Ecampus) instruction in large-enrollment introductory physics courses.

Spring 2026	PH 211 General Physics with Calculus (Studio Lead)
Winter 2026	PH 202 General Physics (Recitation), 170 students
Fall 2025	PH 201 General Physics (Recitation), 150 students
Summer 2025	PH 211 General Physics with Calculus (ECampus), 28 Students
Spring 2025	PH 211 General Physics with Calculus (Studio)
Winter 2025	PH 202 General Physics (Recitation), 210 students
Fall 2024	PH 211 General Physics with Calculus (Recitation), 30 students

Michigan State University

Undergraduate Learning Assistant
Served as primary instructional lead for student groups in a reformed, active-learning introductory physics course, acting as the main point of contact and facilitator of collaborative learning.

Spring 2024	PHY 183B Physics for Science and Engineers I, 65 students
Fall 2023	PHY 183B Physics for Science and Engineers I, 45 students
Spring 2023	PHY 183B Physics for Science and Engineers I, 39 students
Fall 2022	PHY 183B Physics for Science and Engineers I, 40 students

Curriculum Development

Accessibility Focus, PH 211 (Fall 2024)
Implemented new recitation worksheets to improve accessibility, by refining Canvas pages and solutions using accessibility checks (e.g., formatting, color contrast, and screen reader compatibility).

Physics Projects Redesign, PHY 183B (2023-2024)
Reconstructed in-class projects to improve conceptual clarity, by simplifying problem setup and developing clearer diagrams in collaboration with the course instructor.

Mentoring

Graduate Student Mentoring Program, Department of Physics Oregon State University
GTA Mentor Fall 2025
Served as a graduate student mentor in the departmental mentorship program, supporting a first-term graduate student in their transition to being a graduate teaching assistant by meeting weekly to support their teaching and provide resources within the department.

Mutual Mentors Workshop Oregon State University
Participant Summer 2025
Completed a five-week hybrid institute through the Center for Teaching and Learning focused on peer mentoring and classroom observation, developing skills in reflective dialogue, structured conversational techniques, and data-based approaches to observing and supporting teaching practice.

WaMPS Undergraduate-Undergraduate Peer Mentoring Program Michigan State University
Undergraduate Mentor 2023-2024
Served as an upperclassman mentor in a departmental peer mentoring program, supporting newer physics students in navigating the undergraduate physics curriculum and building connections within the department.

Research Experience

Oregon State University – Dept. of Physics Corvallis, OR
Graduate Research Assistant – OSU Physics Education Research March 2025–Present
Advisor: Elizabeth Gire, Ph.D.

- Conducted quantitative analysis of student enrollment and performance data to represent pathways through the physics major at Oregon State University, which features a unique upper-division curriculum: Paradigms in Physics.
- Focused on identifying common course sequences, dropout points, and retake patterns to inform curriculum development and student support strategies.
- Designed and conducted semi-structured interviews with faculty across multiple institutions to investigate approaches, challenges, and attitudes surrounding program assessment practices in higher education physics programs.

Michigan State University – Dept. of Physics and Astronomy East Lansing, MI
Undergraduate Researcher – Physics Education Research Lab September 2022–August 2024
Advisor: Marcos "Danny" Caballero, Ph.D.

- Employed a qualitative interview process to try to understand computation being integrated into high school physics classrooms.
- Explored insights into how the students perceive and interact with computation, as well as the practices employed in the classroom.

Michigan State University – Dept. of Physics and Astronomy East Lansing, MI
Undergraduate Researcher – Physics Education Research Lab March 2022–September 2022
Advisors: Paul Irving, Ph.D. & Daryl McPadden, Ph.D.

- Deployed a card sort interview format to analyze a student's experience, in order to gain insight on how the class is perceived.
- Deduced dissonances in the way this teacher taught physics based on the practice they may be trying to teach, and how a student interacts with those dissonances.

Relevant Coursework

Michigan State University

TE *indicates classes taken in the Department of Teacher Education*

TE 150	Reflections on Learning
TE 101	Social Foundations of Justice and Equity in Education
TE 302	Literacy and Adolescent Learners in School and Community Contexts
PHY 804	Survey of Physics Education Research

Oregon State University

CSSA *indicates courses taken in the College Student Services Administration*
GRAD *indicates courses taken from the Office of Graduate Education*

CSSA 552	Student Development in Universities and Colleges
GRAD 560	Theories of Teaching and Learning in Higher Education
GRAD 561	Course Design and Methods for College and University Teaching
GRAD 607	Capstone Seminar in College and University Teaching
GRAD 610	Internship in College and University Teaching

Presentations

Contributed Talks

1. **J. Rodgers**, P.C. Hamerski, D. McPadden, M.D. Caballero, P.W. Irving, "Computation in High School Physics: How the pieces fit together", *American Association of Physics Teachers Summer Meeting (AAPT)*, Grand Rapids, Michigan. July 2022.

Posters

1. **J. Rodgers**, P.C. Hamerski, D. McPadden, L. A. H. Wood, M.D. Caballero, "Cataloging Engagement In Computational Practices In A High School Physics Classroom", *University Undergraduate Research and Arts Forum (UURAF)*, East Lansing, Michigan. April 2023.

2. **J. Rodgers**, P.C. Hamerski, D. McPadden, M.D. Caballero, P.W. Irving, "How do curriculum design decisions influence student expectations around physics and computation in a computation-integrated physics high school classroom?", *American Association of Physics Teachers Summer Meeting (AAPT)*, Grand Rapids, Michigan. July 2022.

3. **J. Rodgers**, P.C. Hamerski, D. McPadden, M.D. Caballero, P.W. Irving, "How do curriculum design decisions influence student expectations around physics and computation in a computation-integrated physics high school classroom?", *Oslo PER Summer Institute (OPSI)*, Oslo, Norway. June 2022.

Service

- **Member** Coalition of Graduate Employees — 2024–Current

- **Executive Board Member** Michigan State University Special Olympics — 2023–2024
  Worked closely with other members of E-Board to maintain a stream of communication between E-Board and the members of the club.
- **Volunteer** Michigan State University Special Olympics — 2022–2024
  Facilitate individuals with intellectual disabilities in participating in inclusive sports and other activities.
- **Member** Society of Physics Students (SPS) Michigan State Chapter Member — 2021–2024

Technical Experience

Programming

- **Proficient:** Jupyter, Mathematica, Python, R Studio, SQL

Software

- **Proficient:** Microsoft Office, LaTeX, MAXQDA

Awards

- Oregon State University — Graduate Teaching Assistant Award, Honorable Mention 2026
- Michigan State University — Undergraduate Learning Assistant Award 2024